

Insurance Coverage Recommender System

Presentation Speech Script (7 Minutes)

SLIDE 1: Title Slide (30 seconds)

Good morning everyone. Today I'm excited to present our solution to the Generali Data Challenge: an **Insurance Coverage Recommender System** that transforms how we approach cross-selling.

Let me give you a sense of scale. We're working with over **10 million coverage records** from **150,000 real customers** and nearly **2 million insurance quotes**. And the result? We've achieved a **45% hit rate**, meaning nearly half of our customers receive recommendations they actually find valuable.

This isn't just a technical exercise—it's a system ready for production deployment.

SLIDE 2: The Challenge (45 seconds)

So what problem are we solving?

The challenge is **cross-selling insurance coverages** to existing customers. This matters for three critical reasons:

First, **revenue growth**. Targeted recommendations significantly increase policy value per customer.

Second, **customer experience**. Nobody wants random insurance offers cluttering their inbox. Personalized, relevant suggestions actually improve satisfaction.

And third, moving from **intuition to data-driven decisions**. Instead of guessing what customers might need, we're using 10 million records of real customer behavior to make evidence-based recommendations.

Our goal is simple: recommend the right additional coverages to the right customers at the right time.

SLIDE 3: Solution Architecture (60 seconds)

Our solution uses a **multi-model recommender system**—and there's a good reason we chose this approach.

We start with our raw data—10.28 million records with 61 features. Through feature engineering, we distill this down to 21 core predictive features.

Then we run three parallel models:

Collaborative filtering answers the question: "What do customers similar to you typically buy?" It discovers hidden patterns in coverage combinations.

Profile-based machine learning takes a different angle: "Based on your specific characteristics—your age, vehicle, location, risk profile—what coverages make sense for you?"

And finally, our **hybrid combiner** intelligently blends both approaches, giving us the best of both worlds.

Why three models? Because customers are complex. Sometimes behavioral patterns matter most. Sometimes individual characteristics are more predictive. Our hybrid approach captures both signals.

SLIDE 4: Technical Implementation (90 seconds)

Let me dive into the technical approach with more specifics.

Our **collaborative filtering** works like this: we build a co-occurrence matrix showing which coverages are purchased together, then calculate cosine similarity scores between all coverage pairs. For example, when we see that coverage 001—mandatory liability—frequently appears with coverage 057—partial Kasko—we get a similarity score of 0.86, or 86%. This uses sparse matrix optimization to handle our 10 million records efficiently without requiring 100+ gigabytes of RAM.

The **profile-based approach** is quite different. We trained **15 separate Random Forest classifiers**—one for each major coverage type. Each Random Forest contains 100+ decision trees that learn patterns like "if vehicle value exceeds 15,000 euros AND claims equal zero, then probability of buying Kasko is 80%." The forests vote and average their predictions to output probabilities from 0 to 1.

We engineered 21 features across three categories: demographics like age and location, vehicle specifications like brand and commercial value, and behavioral signals like claims history and usage patterns.

The **hybrid combiner** takes both outputs and applies a weighted formula: 60% collaborative score plus 40% profile score. Why this weighting? We tested multiple combinations through cross-validation, and 60-40 performed best. The system also has automatic fallback mechanisms—if a customer has no existing coverages, it uses 100% profile-based recommendations to solve the cold-start problem.

SLIDE 5: Performance Results (60 seconds)

Now let's talk results, because metrics without business meaning don't help anyone.

Our **hit rate of 45%** means that for any given customer, there's nearly a one-in-two chance we recommend something they actually want. That's substantial.

Precision at 10% might sound modest, but in recommender systems, this is industry-standard performance. One in ten recommendations leading to a purchase is exactly what we expect from high-quality systems.

Our **NDCG score of 19%** confirms we're ranking recommendations well—the most relevant suggestions appear at the top.

And we've trained models for **15 of the 27 coverage types**—specifically the most popular ones that represent the bulk of business value.

The bottom line: this system performs at industry standards while handling real-world data complexity.

SLIDE 6: Business Impact & ROI (75 seconds)

Let's translate these metrics into business impact, because that's what matters.

With our 45% hit rate applied to 1.9 million quotes, we're looking at **855,000 potential upsells annually**. Conservative estimates put average policy increases at 15-25% premium uplift. That translates to a **10 million euro annual revenue opportunity**—and that's being conservative.

Beyond revenue, we gain four competitive advantages:

Personalization at scale—individual modeling for every customer, not segment-based guesswork.

Scalability—our architecture efficiently handles millions of records and can grow with the business.

Interpretability—every recommendation is explainable. We can tell agents and customers *why* we're suggesting specific coverages.

And **automation**—we reduce manual underwriting effort, letting agents focus on high-value interactions rather than guessing which products to offer.

The customer experience improves too. These are relevant, non-intrusive recommendations that feel helpful rather than pushy. That drives both satisfaction and revenue.

SLIDE 7: System in Action (45 seconds)

Let me show you a concrete example of how this works in practice.

Imagine a 39-year-old male customer who owns a Fiat Panda and has a clean claims history. He currently has just mandatory liability coverage—the legal minimum.

Our system generates five recommendations, ranked by confidence:

Top recommendation: **Partial Kasko** with a score of 0.8. This makes perfect sense—it's the natural next step from basic liability.

Second: **Theft and Fire coverage** at 0.74. Again, logical for a vehicle owner concerned about protecting their asset.

Third: **Glass coverage** at 0.5, fourth **Natural Events**, and fifth **Roadside Assistance**.

Notice the pattern? These aren't random. They follow a natural progression from basic to comprehensive protection, appropriate for the vehicle type, and matched to a low-risk customer profile. This is what personalization looks like in practice.

SLIDE 8: Conclusion (45 seconds)

So where do we stand? We've delivered a **production-ready system** that handles over 10 million real insurance records with a 45% hit rate and industry-standard precision.

The recommendations are **business-logical**—they make sense in the context of Italian insurance practices. The architecture is **scalable and robust**. And critically, the insights are **interpretable**—stakeholders can understand and trust the recommendations.

For next steps, we recommend **A/B testing** against current recommendation methods, **system integration** with Generali's customer platforms, and **monitoring** to track real-world acceptance rates.

We can also expand with deep learning models, real-time learning from customer feedback, and incorporation of external data sources like weather or economic indicators.

But the core system is ready. It's tested. It's validated. And it's ready to drive value for Generali.

Thank you. I'm happy to take any questions.

TIMING BREAKDOWN

- **Slide 1:** 30 seconds - Introduction and scale
- **Slide 2:** 45 seconds - Problem statement
- **Slide 3:** 60 seconds - Architecture overview
- **Slide 4:** 90 seconds - Technical details (EXPANDED)
- **Slide 5:** 60 seconds - Results and metrics
- **Slide 6:** 75 seconds - Business impact (KEY SLIDE - spend more time here)
- **Slide 7:** 45 seconds - Live example
- **Slide 8:** 45 seconds - Conclusion and next steps

Total: 7.5 minutes (slightly extended for technical detail)

DELIVERY TIPS

1. **Pace yourself:** Don't rush through the technical slides. Pause after key points.
2. **Emphasize business value:** Slides 2, 6, and 8 connect to business outcomes—these are your most important slides.
3. **Use the live example:** Slide 7 makes everything concrete. Reference it when discussing abstract concepts.
4. **Eye contact:** Don't read the slides. They're visual aids for the audience, not your script.
5. **Transitions:** Use clear transitions between slides: "Now that we've seen the challenge, let's talk about our solution..."

- 6. **Questions:** If presenting to executives, they may interrupt with questions. Budget extra time if this is expected.
 - 7. **Energy:** Maintain enthusiasm, especially when discussing business impact. This is exciting work with real revenue potential.
 - 8. **Technical balance:** Gauge your audience. For technical audiences, expand on slide 4. For executives, spend more time on slides 6-7.
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Q&A PREPARATION

Likely Questions:

- "How long to implement?" → 2-3 months for integration, testing, and rollout
- "What about edge cases?" → Hybrid model has fallback mechanisms; cold start uses profile-based only
- "Comparison to current system?" → Need A/B test, but expect 15-20% improvement in conversion
- "Cost to deploy?" → Minimal—uses existing data, models are trained and saved
- "Explainability for customers?" → Feature importance and similarity scores provide clear reasoning
- "Data privacy compliance?" → All data is anonymized, follows GDPR requirements
- "Scalability beyond Italy?" → Architecture is country-agnostic; would need regional training data